

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU is purely text-based MCQ in Indic scripts [Q29] with INDICLID + Unicode filtering for script correctness [Q30] and manual verification per language [Q32]. The deployment is also text-only in Bengali script [WEB-2]. Empirical inspection confirms Bengali items are correctly rendered in Bangla script with no encoding errors [DATASET-D24, D25]. No script-level mismatch exists. The remaining concerns (register/dialect stratification, translation-induced surface form) are content/lexical rather than form properties; the writing-system-level form alignment is strong. One residual concern is that 100% translation rate in the Bengali validation sample [DATASET-D8] may produce translation-flavored surface text that differs subtly from organically Bengali-written prose, but this is at the boundary of form vs. content.

ID	Evidence
Q29	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q30	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q32	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D24	নচিরে প্রশ্নে, দেওয়া চারটি বিকল্পের মধ্যে থেকে সঠিক বিকল্পটি নির্বাচন করুন যা প্রদত্ত শব্দের বিপরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test
D25	নচিরে বাক্যটির সঠিক সক্রিয় রূপ নির্বাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেষ দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেষ সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-2	Bengali alphasyllabary tokenization challenges are documented; Sangraha corpus has ~30B Bengali tokens (https://arxiv.org/abs/2507.23248)

Strengths

- Bengali script (Bangla lipi) rendering is correct and well-formed throughout the sampled split [DATASET-D24, D25]; no script contamination from other Indic scripts observed.
- Documentation explicitly applies INDICLID and Unicode-based filtering to ensure script and language correctness [Q30], plus a final manual sample verification per language [Q32].
- Modality (text) matches the deployment's text-only requirement; no image/audio dependency [Q29].

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D24	নচিরে প্রশ্নে, দেওয়া চারটি বিকল্পের মধ্যে থেকে সঠিক বিকল্পটি নির্বাচন করুন যা প্রদত্ত শব্দের বিপরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test

D25	নচিৰে বাক্যটৰি সঠকি সক্রয়ি ৰূপ নৰিবাচন কৰুন। সবাই তাৰ চমৎকাৰ নৃত্য পৰবিশেণা দ্বাৰা মুগ্ধ হয়ছেলি। ... তাৰ চমৎকাৰ নৃত্য পৰবিশেণা সবাইকে মুগ্ধ কৰছেলি। — Active voice conversion of Bengali sentence — grammar in written Bengali
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal modality (written Bengali script) matches deployment [Q29, WEB-2, DATASET-D24]. No resolution/encoding mismatch — both are Unicode Bengali text.

IF-2: Bangladeshi infrastructure supports Bengali text input/output; urban Dhaka university students have adequate connectivity and devices [WEB-9, WEB-10]. No infrastructure barrier to MILU's input form.

IF-3: MILU excludes reading-comprehension and image-based questions for uniformity [Q29], which slightly narrows form coverage relative to general LLM use, but is acceptable for an MCQ knowledge benchmark.

IF-4: Minor concern: cross-language parallel-translation patterns [DATASET-D7] may produce translationese surface forms in Bengali items, distinct from organically-written Bengali prose; this is a subtle form-quality issue rather than a structural mismatch.

ID	Evidence
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
D7	বাহামান সাম্ৰাজ্যৰে প্ৰথম ৰাজধানী কী ছিলি? ... গুলবাব্ৰ্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D24	নচিৰে প্ৰশ্ননে, দেওয়া চাৰটিকিল্পৰে মধ্য থেকে সেই বকিল্পটিনিৰিবাচন কৰুন যা প্ৰদত্ত শব্দৰে বপিৰীত অৰ্থ প্ৰকাশ কৰে। ... শীতল — Antonym question in Bengali — vocabulary test
WEB-2	Bengali alphasyllabary tokenization challenges are documented; Sangraha corpus has ~30B Bengali tokens (https://arxiv.org/abs/2507.23248)
WEB-9	Urban Dhaka connectivity supports text-based deployment (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-10	https://www.tbsnews.net/bangladesh/over-50-households-use-internet-bangladesh-bbs-1032411

Information Gaps

- Whether translation-induced surface patterns systematically differ from organically-written Bangladeshi Bengali prose at a level that affects model evaluation

Requires Expert Verification

- Native Bangladeshi reader assessment of whether translated Bengali surface form reads naturally